28/09/20

TUT-6

1. $u(r) = \epsilon\left[\left(\frac{r_0}{r}\right)^{12} - 2\left(\frac{r_0}{r}\right)^{6}\right]$

(a) for position of potential minimum —

$$\left.\frac{du}{dr}\right|_{r} = 0$$

$$\Rightarrow \epsilon\left[(-12)\left(\frac{r_0^{12}}{r^{13}}\right) + 12\left(\frac{r_0^{6}}{r^{7}}\right)\right] = 0$$

$$\Rightarrow -12\epsilon\left[\frac{r_0^{12}}{r^{13}} + \frac{r_0^{6}}{r^{7}}\right] = 0$$

$$\Rightarrow \boxed{r = r_0}$$

So, the value of minimum pt. —

$$u(r_0) = \epsilon\left[\left(\frac{r_0}{r_0}\right)^{12} - 2\left(\frac{r_0}{r_0}\right)^{6}\right]$$

$$= \epsilon[1-2] = -\epsilon$$

$$\Rightarrow \boxed{u(r_0) = -\epsilon}$$

(b) Time period of small oscillations around the equilibrium point —

$$T = 2\pi\sqrt{\frac{m}{\left(\frac{d^2u}{dr^2}\right)_{r_0}}}$$

Now $\frac{d^2u}{dr^2} = \epsilon\left[(-12)(-13)\left(\frac{r_0^{12}}{r^{14}}\right) + 12(-7)\left(\frac{r_0^{6}}{r^{8}}\right)\right]$

$$\left.\frac{d^2u}{dr^2}\right|_{r_0} = \epsilon\left[\frac{12\times 13}{r_0^2} - \frac{12\times 7}{r_0^2}\right]$$

$$= \epsilon\,\frac{12\times 6}{r_0^2} = \frac{72\epsilon}{r_0^2}$$

$$\therefore T = 2\pi\sqrt{\frac{m}{72\epsilon/r_0^2}} = 2\pi\sqrt{\frac{r_0^2 m}{72\epsilon}}$$

$$\Rightarrow \boxed{T = \frac{2\pi r_0}{6}\sqrt{\frac{m}{2\epsilon}} = \frac{\pi r_0}{3}\sqrt{\frac{m}{2\epsilon}}}$$

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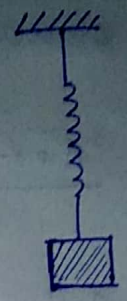
2. $K = 2.05 \times 10^4$ N/m

$m = 10.6$ kg



~~(a)~~ $b = 3$ N.s/m

(a) freq^n of undamped oscillation,

$$\omega_0 = \sqrt{\frac{K}{m}} = \sqrt{\frac{2.05 \times 10^4}{10.6 \text{ kg}}}$$

$$\boxed{\omega_0 = 44/s}$$

freq^n of damped oscillation, $f = \frac{2\pi}{\omega}$ $\frac{\omega}{2\pi}$

$\omega = \cancel{\sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}}$ $\omega = \sqrt{\omega_0^2 - r^2}$ → damping co-efficient

with, $\boxed{r = \frac{b}{2m}}$ $= \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}$

$$= \sqrt{(44)^2 - \left(\frac{3}{2 \times \cancel{2.05 \times 10^4}\ 10.6}\right)^2}$$

≈ 44 rad/s

$\therefore$ ~~$f = \frac{2\pi}{\omega} = \frac{2\pi}{44} = 2 \times \frac{22}{7} \times \frac{1}{44}$~~

$$f = \frac{\omega}{2\pi} = \frac{44}{2\pi} = \frac{44 \times 7}{2 \times 22} = 7 \text{ Hz.}$$

$$\boxed{f = 7 \text{ Hz}}$$

(b) As, the solution for the damped oscillation

$$x = A_0 e^{-rt} \cos(\omega t + \phi)$$

$$x = \underbrace{A_0 e^{-bt/2m}}_{\text{amp. of damped oscillation}} \cos(\omega t + \phi)$$; A_0 → being the amplitude of undamped oscillation

~~change in all~~

So, change in amplitude over a cycle. ($t = T$)

$$\Delta = \frac{A_0 - A_0 e^{-bT/2m}}{A_0}$$

$$= 1 - e^{-bT/2m}$$

$$= \cancel{1 - e^{-}}$$

$$= 0.020$$

$$T = \frac{2\pi}{\omega}$$

$$= \frac{2\pi}{44} = 2 \times \frac{22}{7} \times \frac{1}{44}$$

$$= \frac{1}{7}$$

$$\therefore \frac{bT}{2m} = \gamma T$$

$$= \frac{3 \times 1}{2 \times 10.6 \times 7}$$

$\boxed{\Delta \simeq 20\%}$ ← decreased by 20%.

(c) Let, the initial value of energy $= E_0$ ($E = \frac{1}{2}kx^2$)

Now $E_0 = \frac{1}{2} K A_0^2$

and, for damped system energy

$$E = \frac{1}{2} K A^2 = \frac{1}{2} K \left(A_0 e^{-bt/2m}\right)^2$$

$$= \frac{1}{2} K A_0^2 e^{-bt/m} = E_0 e^{-bt/m}$$

A.T.Q

$$E = \frac{5}{100} E_0$$

$$\Rightarrow E_0 e^{-bt/m} = 0.05 E_0$$

$$\Rightarrow e^{-bt/m} = 0.05$$

$$\Rightarrow \frac{bt}{m} = \ln\left(\frac{1}{0.05}\right)$$

$$\Rightarrow t = \frac{m}{b} \ln\left(\frac{1}{0.05}\right)$$

$$\Rightarrow \boxed{t = 10.6 \text{ sec}}$$

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3. motion of forced damped oscillator –

$$x_a(t) = x_0 \cos(\omega t + \phi)$$

initial con^n : $\left.\begin{array}{l} x(0) = 0 \\ v(0) = 0 \end{array}\right\}$

$x_b(t) \rightarrow$ eq^n of motion of the free damped oscillator.

(a) now eq^n of motion of a forced damped oscillator,

$$\ddot{x}_a + \gamma \dot{x}_a + \omega_0^2 x_a = \frac{F_0}{m} \cos(\omega t) \quad \text{—(1)}$$

with $x_a(t) = x_0 \cos(\omega t + \phi)$

the eq^n of motion for a free damped oscillator,

$$\ddot{x}_b + \gamma \dot{x}_b + \omega_0^2 x_b = 0 \quad \text{—(2)}$$

Now $x(t) = x_a(t) + x_b(t)$

$\Rightarrow \dot{x}(t) = \dot{x}_a(t) + \dot{x}_b(t)$

$\Rightarrow \ddot{x}(t) = \ddot{x}_a(t) + \ddot{x}_b(t)$

Now, ~~for~~ if $x(t)$ is a solution of the driven damped oscillator, then $x(t)$ must satisfy the eq^n (1).

So, $\ddot{x}(t) + \gamma \dot{x}(t) + \omega_0^2 x(t)$

$= (\ddot{x}_a + \ddot{x}_b) + \gamma(\dot{x}_a + \dot{x}_b) + \omega_0^2 (x_a + x_b)$

$= (\ddot{x}_a + \gamma \dot{x}_a + \omega_0^2 x_a) + (\ddot{x}_b + \gamma \dot{x}_b + \omega_0^2 x_b)$

$= \frac{F_0}{m} \cos(\omega t) + 0$ [by (1) and (2)]

$= \frac{F_0}{m} \cos(\omega t)$ So, $x(t)$ satisfies the eq^n of forced HO.

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(b) Let,

$$x_b = X_b e^{-\frac{\gamma t}{2}} \cos(\omega_1 t + \psi) \quad \text{—— 2(a)}$$

Now, we have to find X_b and ω_1 by using initial condition.

In (2a) $\omega_1 = \sqrt{\omega_0^2 - (\gamma/2)^2}$, $\omega_0 = \sqrt{\frac{k}{m}}$

Now

$$x(t) = x_a + x_b$$

$$= X_0 \cos(\omega t + \phi) + X_b e^{-\frac{\gamma t}{2}} \cos(\omega_1 t + \psi) \quad \text{—— (3a)}$$

$$\Rightarrow \dot{x} = -\omega X_0 \sin(\omega t + \phi) + X_b \left[-\frac{\gamma}{2} e^{-\frac{\gamma t}{2}} \cos(\omega_1 t + \psi) - \omega_1 e^{-\gamma t/2} \sin(\omega_1 t + \psi) \right]$$

$$= -\omega X_0 \sin(\omega t + \phi) + X_b e^{-\frac{\gamma t}{2}} \left[-\frac{\gamma}{2} \cos(\omega_1 t + \psi) - \omega_1 \sin(\omega_1 t + \psi) \right] \quad \text{—— (3b)}$$

Now initial condition:

at $t = 0$ $x = 0$

$$\therefore 0 = X_0 \cos\phi + X_b \cos\psi \quad \text{—— (4a)}$$

at $t = 0$, $v = 0$

$$\therefore 0 = -\omega X_0 \sin\phi + X_b \left[-\frac{\gamma}{2} \cos\psi - \omega_1 \sin\psi \right]$$

$$= -\omega X_0 \sin\phi - X_b \left[\frac{\gamma}{2} \cos\psi + \omega_1 \sin\psi \right] \quad \text{—— (4b)}$$

For solving Let,

$$c = X_b \cos\psi$$

$$s = X_b \sin\psi$$

So, (4a) —— $c = -X_0 \cos\phi$

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and by (4b) –

$$-\omega x_0 \sin\phi - \frac{\gamma}{2} c - \omega_1 s = 0$$

$$\Rightarrow s = \frac{x_0}{\omega_1}\left[-\omega \sin\phi + \frac{\gamma}{2}\cos\phi\right]$$

so, using 'c' & 's' eqn (3a) & (3b) can be written as,

$$x = x_0 \cos(\omega t + \phi) + e^{-\frac{\gamma t}{2}}\left[c\cos(\omega_1 t) - s\sin(\omega_1 t)\right] \quad \text{—(5a)}$$

~~at t=0~~

$$\cancel{x(t=0) = x_0 \cos\phi + [c]}$$

$$\cancel{= x_0 \cos\phi}$$

and $\dot{x} = -\omega x_0 \sin(\omega t + \phi) - e^{-\frac{\gamma t}{2}}\left[\left(\frac{\gamma}{2}\right) c \cos(\omega_1 t) - \left(\frac{\gamma}{2}\right) s \sin(\omega_1 t) + \omega_1 c \sin(\omega_1 t) + \omega_1 s \cos(\omega_1 t)\right]$

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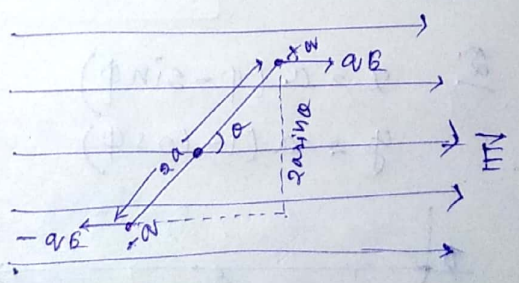
1. Consider a dipole oscillating in e.f (E) as shown in figure.



The e.f will ~~cause~~ try to orient the dipole at various angles.

when, it is at an angle 'θ' with the field,

The torque acting on the dipole is –

$$\vec{\tau} = \vec{r} \times \vec{F} = rF \sin\theta$$

$$\Rightarrow \tau = (2a)(qE)\sin\theta$$

$$= 2qaE \sin\theta$$

$$\Rightarrow \tau = pE \sin\theta$$

with,

$p = q(2a)$

$= 2aq$ (dipole moment of the system)

Now, this torque acts as a restoring force over the system.

i.e. It will try to orient the system and τ will try to bring the system in its original position.

So, the eqn of motion for the system,

$$I\alpha = -PE\sin\theta$$

α → angular acc.

$-PE\sin\theta$ → as restoring force

I → MI of the system.

$$\Rightarrow I\cdot\frac{d^2\theta}{dt^2} + PE\,\theta = 0 \quad \text{, for small angles, } \sin\theta \simeq \theta$$

$$\Rightarrow \frac{d^2\theta}{dt^2} + \frac{PE}{I}\theta = 0$$

$$\Rightarrow \boxed{\frac{d^2\theta}{dt^2} + \omega^2\theta = 0} \rightarrow \text{SHM}$$

Where, $\omega = \sqrt{\frac{PE}{I}}$

So, $\boxed{T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{I}{PE}}}$

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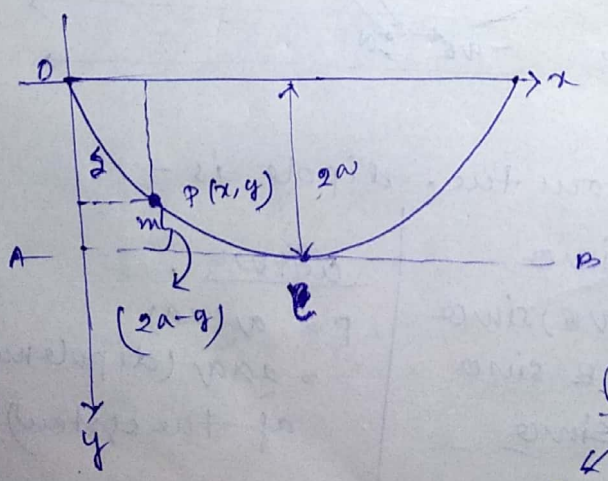
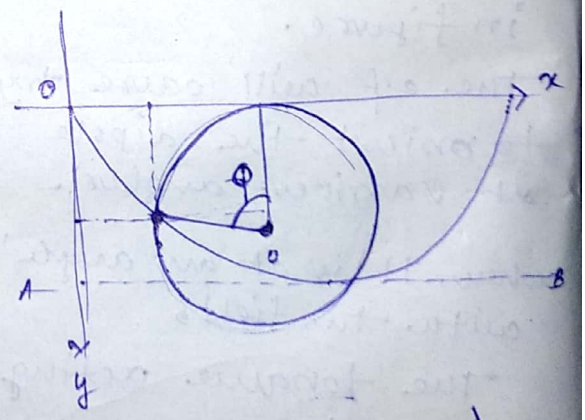
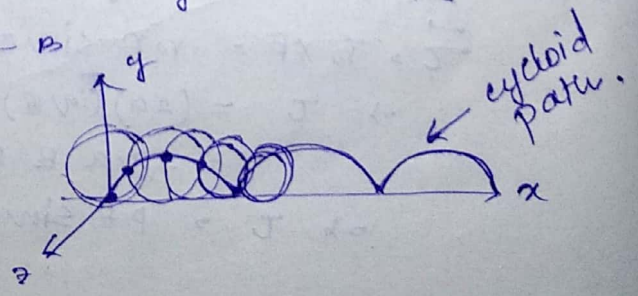
2. $x = a(\phi - \sin\phi)$

$y = a(1 - \cos\phi)$







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(a) Let, P be the position of the bead at any instant t.

and, let 's' be the arclength (upto P) along the cycloid at any point 'O'.

By conservation of energy, relative to line AB through the minimum point of the cycloid, —

$$(PE + KE)_P = (PE + KE)_O$$

$$\Rightarrow mg(2a - y) + \frac{1}{2}m\left(\frac{ds}{dt}\right)^2 = mg(2a) + 0 \quad \left(\text{at } O,\ v = 0\right)$$

$$\Rightarrow v^2 = \left(\frac{ds}{dt}\right)^2 = 2gy \quad \text{—— (1)}$$

$$\Rightarrow \boxed{v = \sqrt{2gy}} \rightarrow \text{for any arbitrary point.}$$

So, speed at bottom of the path, i.e at $y = 2a$ (at C)

$$\boxed{v = \sqrt{2g \cdot 2a} = 2\sqrt{ga}}$$

(b) Now

from (1) — $v^2 = \left(\frac{ds}{dt}\right)^2 = 2gy$

now, for small arc length,

$$ds^2 = dx^2 + dy^2$$

$$\Rightarrow \left(\frac{ds}{dt}\right)^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$$

$x = a(\varphi - \sin\varphi)$

$\dot{x} = a(\dot{\varphi} - \cos\varphi\,\dot{\varphi})$

$y = a(1 - \cos\varphi)$

$\dot{y} = +a\sin\varphi\,\dot{\varphi}$

$$= a^2(\dot{\varphi} - \cos\varphi\,\dot{\varphi})^2 + a^2\sin^2\varphi\,\dot{\varphi}^2$$

$$= 2a^2(1 - \cos\varphi)\,\dot{\varphi}^2$$

$$\Rightarrow 2gy = 2a^2(1 - \cos\varphi)\,\dot{\varphi}^2$$

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$\Rightarrow 2ga(1-\cos\varphi) = 2a^2(1-\cos\varphi)\dot{\varphi}^2$

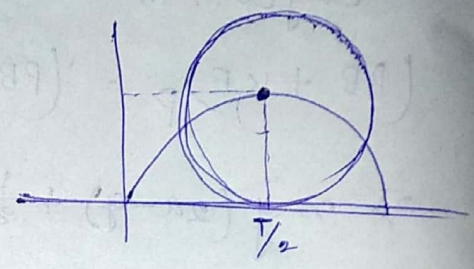
$\Rightarrow \dot{\varphi}^2 = g/a$

<u>BC</u> at $\varphi = 0 \Rightarrow t = 0$

at $\varphi = 2\pi \Rightarrow t = T/2$, $T \rightarrow$ time period.

so, $\dot{\varphi} = \sqrt{g/a}$

$\Rightarrow \varphi = \sqrt{\frac{g}{a}}\, t + c_1$



at $\varphi = 0 \Rightarrow t = 0$, $\Rightarrow c_1 = 0$

$\therefore \varphi = \sqrt{\frac{g}{a}}\, t$

at $\varphi = 2\pi$, $t = T/2$

$\therefore 2\pi = \sqrt{\frac{g}{a}} \cdot \frac{T}{2}$

$\Rightarrow T = 4\pi\sqrt{a/g}$

$\Rightarrow \boxed{T = 2\pi\sqrt{4a/g}}$ — (a)

for simple pendulum, $T = 2\pi\sqrt{\frac{l}{g}}$

so, the time period of bead is same as that of a pendulum of length $4a$.

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